

Vol 14 Chapter 2 — Reed-Solomon Coding on GF(128)

Keeper (author pass — deep math/physics revision)

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Chapter 2 — Reed-Solomon Coding on GF(128)

Level 1 — one sentence

Reed-Solomon codes — block codes over finite field $GF(2^g) = GF(128)$ with the BST primary $g = 7$ selecting the natural alphabet size — provide optimal Maximum Distance Separable (MDS) error correction up to $\lfloor (n - k)/2 \rfloor$ symbol errors per codeword, and BST identifies this as the substrate's native error-correction mechanism (Paper #122).

Level 2 — graduate-physicist precision

2.1 $GF(2^g) = GF(128)$

Finite field $GF(2^7) = GF(128)$: 128 elements, characteristic 2, dimension 7 over $GF(2)$.

Constructed: $GF(128) \cong GF(2)[x]/(p(x))$ where $p(x)$ is an irreducible degree-7 polynomial over $GF(2)$.

Multiplicative group $GF(128)^* \cong \mathbb{Z}/127$ (cyclic of order $128 - 1 = 127 = N_{\max} - 10$, prime).

2.2 Reed-Solomon RS(n, k)

Encode message $m = (m_0, m_1, \dots, m_{k-1}) \in GF(128)^k$ as codeword $c \in GF(128)^n$ with $n \leq 127$.

Singleton bound: minimum distance $d \leq n - k + 1$. RS codes achieve equality (MDS).

Error correction: can correct any $t = \lfloor (n - k)/2 \rfloor$ symbol errors.

2.3 Standard parameters

Common RS choices: - RS(255, 223) over GF(256): used in NASA Voyager, Compact Disc audio - RS(127, k) over GF(128): BST-natural - RS(n, k) with $n - k = 2t$ for t -error correction

2.4 BST primary anchoring

- $g = 7 \rightarrow$ field $GF(2^g) = GF(128)$ — substrate alphabet
- Block length $n \leq 127 = N_{\max} - 10$ — close to substrate ceiling
- $127 = M_g$ (g -th Mersenne prime $= 2^7 - 1$) — Vol 11 Ch 10 substrate Mersenne ladder
- Cyclic group $\mathbb{Z}/127$: substrate's natural cyclic structure

2.5 Substrate implementation hypothesis

Hypothesis (Paper #122): substrate implements RS-like coding on GF(128) at the Koons-tick rate, providing error correction against substrate-level noise.

Evidence: - Genetic code (Vol 13 Ch 2): codon-amino acid redundancy $64 \rightarrow 20$ is RS-like - Cosmological information bounds (Vol 4): consistent with substrate-RS - BST primary $g = 7$ selects $GF(128)$ uniquely

Falsifier: identify a different finite field implementing BST observables better → would refute $g = 7$ as the substrate-natural alphabet exponent.

2.6 K-audit anchors

- K59: cyclotomic mechanism framework
- K68: RS computation
- Paper #122: Information Substrate

Level 3 — 5th-grader accessibility

Reed-Solomon codes are how your phone, CD, satellite communications correct errors. Use a finite field $GF(128)$ — 128 numbers with addition and multiplication that wrap around. Add redundancy: send 127 symbols to encode 100, can correct up to $(127 - 100)/2 = 13$ errors. In BST: $128 = 2^g$ where $g = 7$ is a BST primary; substrate uses RS-coding on $GF(128)$ at the Koons-tick rate. Genetic code uses similar redundancy (64 codons → 20 amino acids).

What comes next

Chapter 3 develops Shannon channel capacity.

Where to look this up

- Reed-Solomon 1960
- MacWilliams-Sloane, *Theory of Error-Correcting Codes*
- BST: K59, K68, Paper #122